

QUANTUM STATE REFERENCE: THE UNIFIED MARKET PHYSICS THEORIES

TARGET AUDIENCE: Claude Code (Development AI) **PURPOSE:** You already possess the raw mathematical implementation in `quantum_field_engine.py` (919 lines). This document provides the **Design Intent, State Mappings, Decision Rules, and Architectural History** necessary to wire those quantum state computations back to the execution, scoring, and gating logic.

This reference bridges the gap between the theoretical physics and the deterministic trading behavior of the Bayesian-AI system.

1. THEORY I: THE NIGHTMARE PROTOCOL (The Gravitational Field)

1.1 Mathematical Formalization

- **The Ornstein-Uhlenbeck (O-U) Process:** Defines the baseline mean-reverting behavior of the particle.

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t$$

- **Gravitational Force ($F_{gravity}$):** Pulls the particle back to the mean.

$$F_{gravity} = -\theta \cdot Z_t \cdot \sigma$$

- **Boundary Repulsion Force ($F_{repulsion}$):** Electrostatic-like pushback emitted by the Event Horizon.

$$F_{repulsion} = \frac{k}{r^3} \quad \text{where} \quad r = 3 - |Z_t|$$

- **Roche Limit (2σ):** Boundary where tidal forces tear the localized trend apart.
- **Event Horizon (3σ):** Absolute singularity boundary; probability density $\rightarrow 0$.

1.2 Design Intent

The market is not a random walk; it is a liquidity-driven gravitational well. Market makers algorithmically pull price back to VWAP/Mean (μ). The Standard Error bands (σ) are not mere statistical metrics—they represent structural physical boundaries (Roche Limits).

- **What it models:** Localized mean reversion, inventory rebalancing, and the exhaustion of kinetic order flow against limit-order walls.
- **What breaks if ignored:** A standard trend-following bot will buy a breakout at the exact moment the micro-trend hits a macro 3σ wall, getting crushed by the "rubber band" snapback.

1.3 State Variables

- `z_score` : Spatial coordinate of the particle.
- `micro_velocity` : First derivative of price (dP/dt).
- `F_gravity` , `F_repulsion` , `net_force` : Computed magnitudes of forces.
- `prob_center` , `prob_upper` , `prob_lower` : The Boltzmann probabilities derived from potential energy of the Net Force.

1.4 Decision Rules

- **ENTER (Wave Function Collapse):** Price has pulled back to a stable gravitational zone ($-1.0 \leq Z \leq 1.0$), and the computed Net Force strongly biases toward a specific target band. The particle is "cleared" to travel.
- **STAY FLAT (Event Horizon):** Price is near the singularity ($|Z| \geq 2.5$). Trend-continuation entries are hard-blocked. You do not initiate momentum when the particle is out of bounds.
- **EXIT (Roche Limit Hit / Death Hook):** Price touches 2σ AND `micro_velocity` flips sign. The localized kinetic energy has hit the wall and died. Exit immediately.

2. THEORY II: THE THREE-BODY PROBLEM (Fractal Chaos)

2.1 Mathematical Formalization

- The Superposition Wave Function (Ψ):

$$\Psi = a_0 \Psi_{center} + a_1 \Psi_{upper} + a_2 \Psi_{lower}$$

- Boltzmann Probabilities (P_i):

$$P_i = \frac{\exp(-\beta E_i)}{\sum_j \exp(-\beta E_j)}$$

- Shannon Entropy (S):

$$S = - \sum_i P_i \ln(P_i)$$

- **Fractal Coherence (C):** Dot product (alignment) of the Micro, Meso, and Macro directional vectors.

2.2 Design Intent

In astrophysics, three interacting bodies create deterministic chaos. In trading, these are the Macro (1h), Meso (5m), and Micro (15s) timeframes. A trader executing solely on a 15-second chart is blind to the 1-hour gravitational pull that will inevitably override their position.

- **What it models:** The absolute necessity of multi-timeframe phase alignment. We model chaos using Shannon Entropy to find brief, probabilistic windows of stability.
- **What breaks if ignored:** "Death by a thousand cuts." The bot takes perfectly valid micro-setups that are immediately vaporized by conflicting macro order flow.

2.3 State Variables

- `entropy` : The Shannon entropy score of the current bar's wave function.
- `coherence` : The calculated alignment [0.0 to 1.0] of the three timeframes.

2.4 Decision Rules

- **TRADEABLE (Alignment/Resolution):** * `entropy` is LOW (meaning the wave function has collapsed and one directional probability is dominant, e.g., $P_{upper} > 0.85$).
 - `coherence` is HIGH (Macro, Meso, Micro vectors are mathematically synchronized).
- **UNTRADEABLE (Chaos/Misalignment):** * `entropy` is HIGH (probabilities are flat/uncertain, e.g., $[0.33, 0.33, 0.33]$).
 - `coherence` is LOW. The system completely halts entries.
- **EXIT (Decoherence):** A previously high `coherence` state rapidly decays into high `entropy` while a trade is open.

3. THEORY III: THE RESONANCE CASCADE (The Alpha Event)

3.1 Mathematical Formalization

- **The Lyapunov Exponent (λ):** Measures the rate of separation of infinitesimally close trajectories. Determines stability ($\lambda < 0$) vs. cascade ($\lambda > 0$).

$$|\delta Z(t)| \approx e^{\lambda t} |\delta Z(0)|$$

- **Hurst Exponent (H):** Measures the long-term memory of the series.

3.2 Design Intent

Occasionally, the O-U mean-reverting physics break down. When the frequencies of the Three Bodies perfectly synchronize, constructive interference shatters the Roche Limits. The market transitions from stable to chaotic, resulting in a Black Swan or a massive trend day.

- **What it models:** Liquidity cascades, short squeezes, and algorithmic momentum ignition.
- **What breaks if ignored:** The bot will repeatedly try to fade/short a massive breakout because the price hit 3σ , effectively stepping in front of a freight train and blowing up the account.

3.3 State Variables

- `lyapunov_exponent` : Real-time calculation of λ .
- `macro_hurst_exponent` : Rolling Hurst calculation on the Macro timeframe.
- `is_resonance_cascade` : Boolean flag activating the Cascade Protocol.

3.4 Cascade Mode Protocol Details

When `is_resonance_cascade = True`, standard O-U physics are suspended.

- **Exact Trigger Conditions:**
 1. `macro_hurst_exponent` > 0.55 (Mathematical proof of persistent memory).
 2. `lyapunov_exponent` > 0 (Mathematical proof of chaotic exponential separation).
 3. `coherence` > 0.90 (Cross-TF vector synchronization).
 4. Macro Volatility (σ_{macro}) is visibly expanding directionally.
- **Suppressed Exits:**
 - Standard Take Profit (Do not arbitrarily cap gains in a Black Swan).
 - Band Exits / Envelope (Do not fade the Roche limits, as they are actively breaking).
- **Active Exits:**
 - `survival_stop` (If Bayesian win probability decays below 50% despite the cascade).
 - `tidal_wave` (If volatility expands violently *against* our position).
- **Termination Conditions (Reverting to Nightmare Protocol):**
 - `macro_hurst_exponent` decays back toward 0.50 (Random walk returning).
 - Macro DMI/ADX vector hooks downward violently (Order flow exhaustion).
 - `lyapunov_exponent` flips negative (Trajectory stabilizing).

4. THE ORIGINAL SCORING LOGIC (Pre-Gate Cascade)

Before the rigid boolean screening gates (`tools/analyze_gates.py` , `screening_gates.json`) were introduced in the newer architecture, the system relied purely on continuous quantum probabilistic resolution.

If you need to reconstruct the pure scoring loop, the logic flowed as follows:

1. **Force to Energy:** The `quantum_field_engine` computed $F_{gravity}$, $F_{repulsion}$, and $F_{momentum}$, summing them into a Net Force.
2. **Energy to Probability:** Net Force was converted to potential energy states, which were then converted into Boltzmann weights (`prob_center` , `prob_upper` , `prob_lower`).
3. **Tunneling Computation:** The system ran a Monte Carlo O-U simulation to find the `tunnel_probability` —the statistical likelihood that the particle could reach the target band without hitting the stop-loss boundary first.
4. **Execution Decision:** A trade was executed purely on mathematical edge:
 - Highest wave function probability (P_i) was > 0.75
 - AND `tunnel_probability` was > 0.60
 - AND `entropy` was low.
 - *Note: There were no hard boolean "gates"; it was a continuous spectrum of expected value.*

5. UNIMPLEMENTED / THEORETICAL CONCEPTS

These concepts were heavily researched and mapped mathematically but were **never fully implemented** in the live execution code. If you encounter references to them in the codebase or notes, treat them as theoretical scaffolding:

1. **Continuous Phase Transitions:** The current execution engine treats the shift from *Nightmare Protocol* (Mean Reversion) to *Resonance Cascade* (Trend) as a discrete boolean switch. The original mathematical intent was a continuous phase transition (like water to steam), where strategy weights would smoothly interpolate as the Hurst Exponent slid from 0.49 to 0.51.
2. **Cascade-Specific Trailing Mathematics:** A dynamically calculated trailing stop directly proportional to the inverse of the Lyapunov Exponent ($1/\lambda$). As chaotic separation accelerated, the stop would loosen to allow for volatility; as λ slowed, the stop would tighten aggressively to capture the peak.
3. **True Cross-TF Wave Interference:** The current system uses a simplified `coherence` dot-product to measure alignment between the 3 timeframes. The unimplemented theory